

AIP Batch Export (CPP-006)

CPP-Identifier	CPP-006
CPP-Label	AIP Batch Export
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Change history	Comments
Version 1.0.0 - 2025-08-29	Milestone version
Version 1.1.0 - 2026-03-26	Migration to XML
Version 1.2.0 - 2026-07-24	Added a step group for steps specific to a single AIP, steps renumbered accordingly.

1. Description of the CPP

As part of an exit strategy, the TDA batch exports Information packages and all associated Metadata in a manageable format/structure for Ingest into another TDA.

Inputs and outputs

Input(s)	
Data	AIP(s)
	File(s)
Metadata	Storage Metadata Information (Identifier of AIP to be exported, including to AIP versions)
	Target location for export
Documentation/ guidance	Contractual Documentation for batch export (for service providers, if applicable)
	AIP specification
	Exit Scenario Plan
	Documentation of Export AIP (supported target structure and format)
Output(s)	
Data	External copies of one or several complete AIPs
Metadata	Only Metadata contained in AIPs

Definition and scope

While the TDA manages *Information packages* across their life cycle, stores them as *AIPs*, and delivers them as *DIPs* to its consumer, it also needs to plan for situations in which an export of *AIPs* becomes necessary.

Scenarios that require an AIP Batch Export include, but are not limited to:

- *Exit Scenario*: The TDA changes a major part of its system infrastructure, such as the digital preservation workflow software or the database.
- *Succession Planning*: The TDA ceases to exist and needs to hand over its archival holdings to a successor.
- *Additional External Storage*: The TDA wants to store additional offline copies of their *AIPs*. The *AIP* export will create a *physical AIP* as opposed to a logical one. This includes e.g. *Metadata* which might be kept in a database to be written into an XML or JSON *File*.
- *Disposal*: If the TDA decides to dispose data (e.g. as part of Re-Appraisal) the data might be handed over to an external entity such as a different TDA that wants to preserve it. The *AIP* likely contains a level of information richness (e.g. in form of versions and full audit trails) that the *DIP* format of the TDA does not offer.
- *External Request for Export*: An external entity (e.g. a customer for a digital preservation service) might request the data preserved by the TDA for use cases such as verification of provenance information.

This CPP may export 1-n *AIPs* including all *AIP* versions from the TDA to a storage location external to the archival storage. The target location can be located externally (e.g. an external SFTP server) or internally (e.g. to a different network share). The process makes no assumption as to the type of storage exported to (e.g. disk, tape).

The exported *AIP* should include all available information where possible (i.e. contain all *Files* and all accompanying *Metadata*). In addition to *Descriptive*, *Structural*, *Administrative* and *Technical Metadata*, this includes any *Provenance metadata* which might exist and describes any events undertaken on the *Object* during its lifespan within the TDA.

The format and structure of the exported *AIP* should be documented including an explanation of all internal identifiers or schemas that might be used within. This documentation needs to be provided as a sidecar *File* with the AIP Batch Export.

Process description

Trigger event(s)

Trigger Event	CPP-identifier
AIP is requested from external entity (e.g. customer for digital preservation service)	
Succession plans are applied to transfer custody of <i>Objects</i> to another TDA	
The TDA extends bit-level preservation capabilities and creates new (internal or outsourced) copies of <i>AIPs</i> (e.g. to implement geographically distributed storage or to copy <i>AIPs</i> to an offline storage system)	
The TDA changes its system and all <i>Information packages</i> need to be exported (i.e. Exit scenario)	
The TDA has flagged data for disposal and the <i>AIPs</i> are to be handed over to an external entity	

Step-by-step description

No	Supplier	Input	Steps	Output	Customer
↓ sequence					
1			Receive list of <i>AIPs</i> to be exported		
2		Target structure and format of exported <i>AIP</i>	<i>Optional:</i> If multiple target structures and formats are supported: Set export to transform exported <i>AIP</i> to desired structure and format		
3		List of <i>AIPs</i> to be exported	Select <i>AIPs</i> to be exported in TDA (e.g. by creating a set based on AIP identifiers and <i>AIP</i> version identifiers, if applicable)	Inventory of <i>AIPs</i>	
				Inventory of <i>AIP</i> versions	

No	Supplier	Input	Steps	Output	Customer
				Location of information belonging to <i>AIPs</i> and <i>AIP</i> versions	
4		Inventory of <i>AIPs</i>	<i>Optional:</i> Depending on handling method for updates in TDA: For each <i>AIP</i> : lock <i>AIP</i> to avoid it being updated during export	Locked location of source <i>AIP</i>	
		Inventory of <i>AIPs</i> versions			
5	↓ sequence - For each <i>AIP</i> in inventory list: Copy <i>AIP</i> to target location.				
5.1		Inventory of <i>AIPs</i>	Create logical structure for <i>AIP</i>		
		Inventory of <i>AIPs</i> versions			
		<i>Storage Metadata Information</i>			
5.2			Write logical structure to target location	Logical structure tree of exported <i>AIPs</i>	
5.3			Copy <i>Files</i> to target location	Files in logical structure tree of exported <i>AIPs</i>	
5.4		Database query for <i>AIP</i>	If (additional) <i>Metadata</i> is stored in other place tha <i>AIP</i> (e.g., database): collect all other information belonging to <i>AIP</i> (<i>Metadata</i>)	All information belonging to <i>AIP</i>	
5.5		Result of Database query for <i>AIP</i>	If (additional) <i>Metadata</i> is stored in database, transform <i>Metadata</i> into target format and write to target location, add export as provenance information into <i>Metadata File</i> during transformation	Metadata <i>File(s)</i> in target location	
		Mapping for export of selected <i>Metadata</i> from database to flat <i>Metadata</i>		Provenance information in metadata <i>File(s)</i>	

No		Supplier	Input	Steps	Output	Customer
5.6	5.6		Inventory of <i>AIPs</i>	The <i>AIP</i> in the target should be verified, including <i>Files</i> and <i>Metadata</i> to ensure it is complete and correct ' (e.g. using checksums)	Number of <i>Files</i> matches (step 5.8)	
			Inventory of <i>AIP</i> versions		All checksums match (step 5.8)	
			List of <i>Files</i> (excluding <i>Metadata</i> files) at target location		Alert that number of <i>Files</i> do not match	
			<i>Fixity Metadata</i> from source <i>AIP</i>		Alert that any of the <i>File</i> checksums does not match	
	5.7		<i>AIP</i> ID	<i>Optional:</i> Depending on handling method for updates in TDA: Release lock on <i>AIP</i> .	Released lock	
	5.8		<i>Files</i> in target location	<i>Optional:</i> If required: Perform additional repacking of <i>Files</i> at target location to meet target structure and format of exported <i>AIP</i> (e.g., tarball)		
			Target structure and format of exported <i>AIP</i>			
6			Documentation of Export <i>AIP</i> (supported target structure and format of exported <i>AIP</i>)	Add documentation to target location	Documentation of Export <i>AIP</i> (supported target structure and format of exported <i>AIP</i>)	
7			Inventory of <i>AIPs</i>	Document export in audit trail	Audit trail for export	

Rationale(s) and worst case(s)

Rationale	Impact of inaction or failure of the process
When a TDA is closed down, it will have to pass data on to another TDA as part of succession planning. This might have to happen on short notice.	Data is lost.
When a TDA acts as a service provider, data will have to be exported in bulk to those who are subscribing to the service.	Data owner potentially loses control over data.
If a TDA is using external services and systems, e.g. for storing and preserving its <i>AIPs</i> , then it may need to exchange or transfer <i>AIPs</i> to or from third-party services and systems.	Data is lost, data can't be efficiently transferred between TDAs and preservation services.

2. Dependencies and relationships with other CPPs

Dependencies

CPP-ID	CPP-Title	Relationship description
CPP-001	Checksum Generation and Recording	<i>Fixity metadata</i> is used to verify the integrity of data written into the exported <i>AIP</i> .
CPP-002	Checksum Validation	To ensure the integrity of the data during transport from the TDA storage, the exported <i>Files'</i> checksums need to be verified.

Other relations

Relation	CPP-ID	CPP-Title	Relationship description
Not to be confused with	CPP-017	Disposal	By default, batch export does not remove the content from the TDA.
Not to be confused with	CPP-025	Enabling Access	Access is typically granted to the <i>DIP</i> which may be different from the <i>AIP</i> (e.g. the <i>DIP</i> may only present the last version or one of several <i>Representations</i>). The <i>AIP</i> contains all preservation <i>Metadata</i> which a <i>DIP</i> may not.
Affected by	CPP-021	AIP Versioning	Versioning impacts how the export will have to be run and where and how information about the versions may be found. In addition a policy might determine if only the last or all versions should be exported.
Affinity with	CPP-011	Replication	Replication creates new parallel copies of <i>AIPs</i> within a TDAs archival storage. AIP Batch Export exports <i>AIPs</i> to external locations.

3. Links to frameworks

Certification

Certification framework	Term used in framework to refer to the CPP	Section
CTS	"Technical aspects of business continuity, disaster recovery and succession planning"	R15 Technical Infrastructure Organisational aspects are covered in R03 Continuity of Service
Nestor Seal	Export	C12 Crisis / successorship management
ISO 16363	Export	4.3.5 Also touched in 3.1.2.2 "The repository shall monitor its organizational environment to determine when to execute its succession plan, contingency plans, and/or escrow arrangements."

Other frameworks and reference documents

Reference Document	Term used in framework to refer to the process	Section
OAIS	/	The process is not described in OAIS but one of its purposes, Succession Planning, is briefly mentioned in 3.3.6 "Follows established preservation policies and procedures".
PREMIS	/	/

4. Reference implementations

Publicly available documentation

Institution	Organisation type	Language	Hyperlink
TIB – Leibniz Information Centre for Science and Technology and University Library, DE	National library; Non-commercial digital preservation service; Research infrastructure; Research performing organisation	English	https://wiki.tib.eu/confluence/spaces/lza/pages/93608980/Export+and+exit+scenario
CSC – IT Center for Science Ltd., FI	Non-commercial digital preservation service	Finnish	https://urn.fi/urn:nbn:fi-fe2024051731943 (section 13)
Archivematica, CA	Digital preservation system	English	https://www.archivematica.org/en/docs/archivematica-1.17/user-manual/archival-storage/archival-storage/#downloading-an-aip